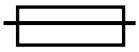
11 PLUS PREPARATION

NON-VERBAL REASONING PAPER

Name:	
Date of Birth:	
Today's Date:	

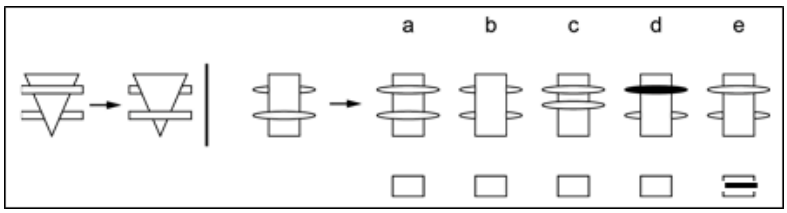
READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING.

- You have 15 minutes to complete the paper.
- Mark your answer like this: 
- This paper features 13 different question types. An example is given at the start of each question type.
- Each question is worth one mark. If you can't do a question, then move on.
- If you finish early check your answers for mistakes.
- Once the test has started you may not ask for help.

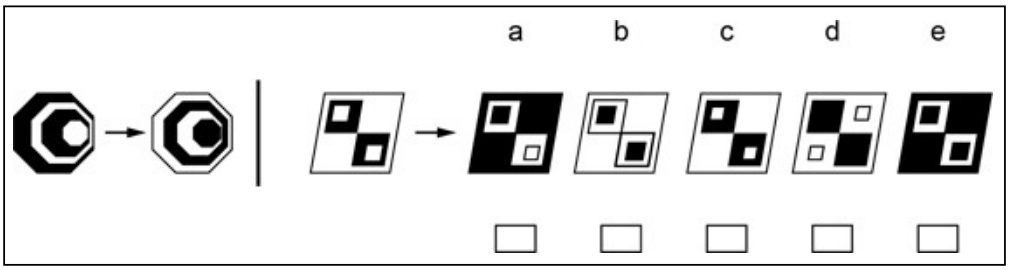
1. LIKE SHAPES / TYPE 1

Work out which of the answer shapes is related to the question shape in the same way that the example shapes are related.

Example



1



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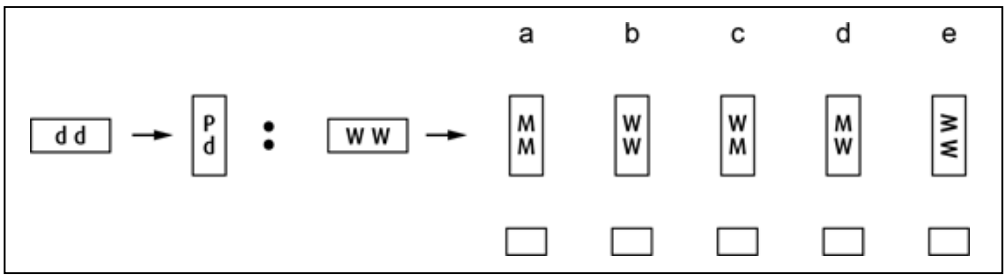

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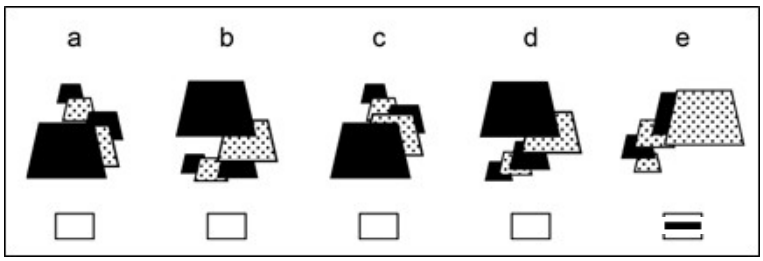
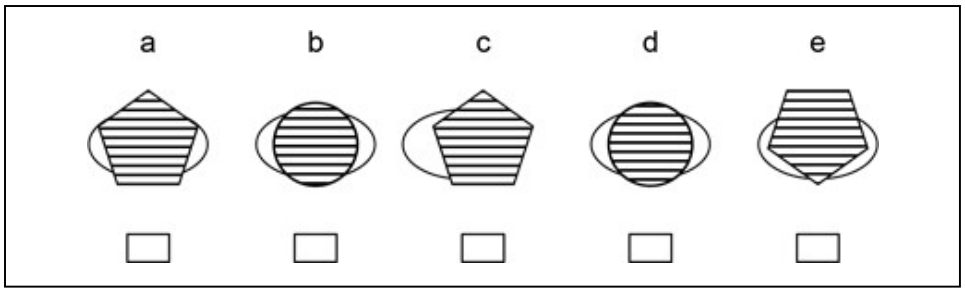
2. ODD ONE OUT / TYPE 2

Work out which of the shapes is different to the rest.

Example



1



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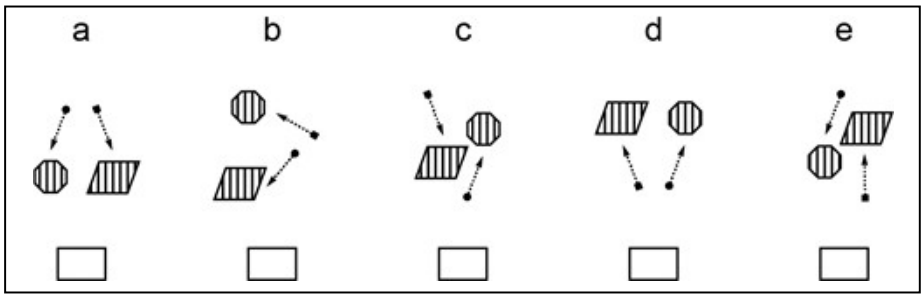
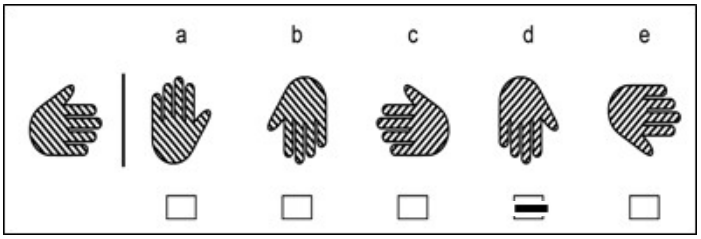


3. ROTATIONS / TYPE 3

Work out which of the answers is a rotation of the question shape.

Example



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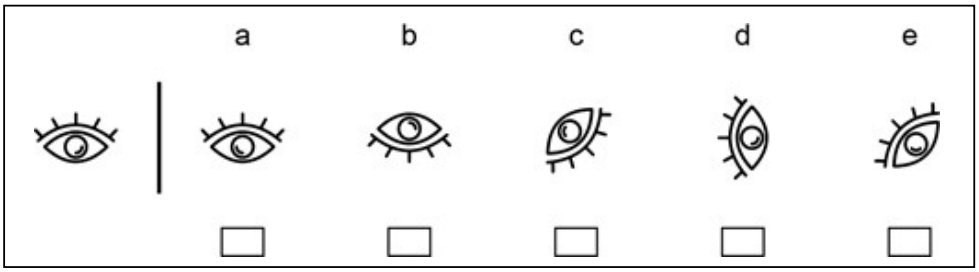

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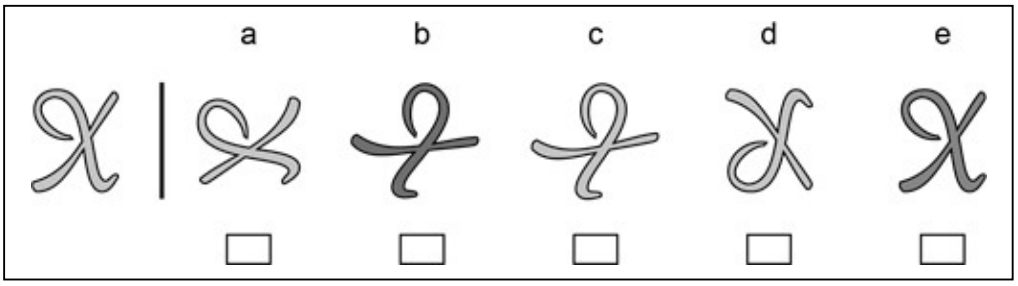

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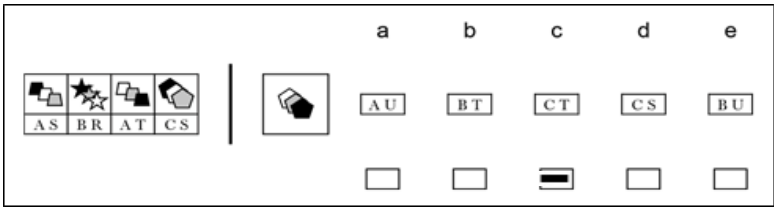
4. CODE BREAKER / TYPE 4

Look at the sequence to work out how the codes match the shapes, then select the correct answer code to match the question shape.

Example



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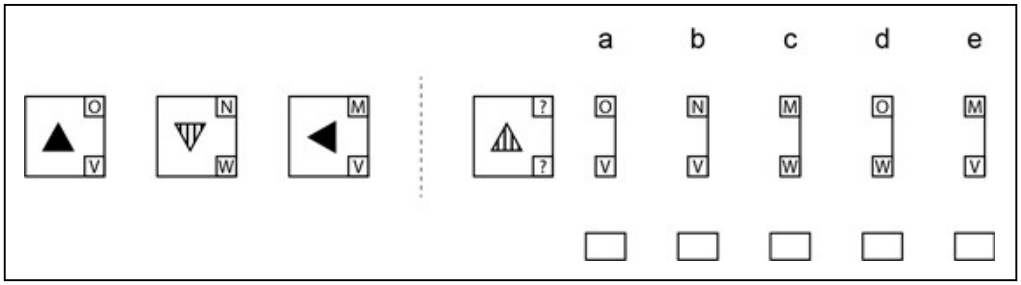

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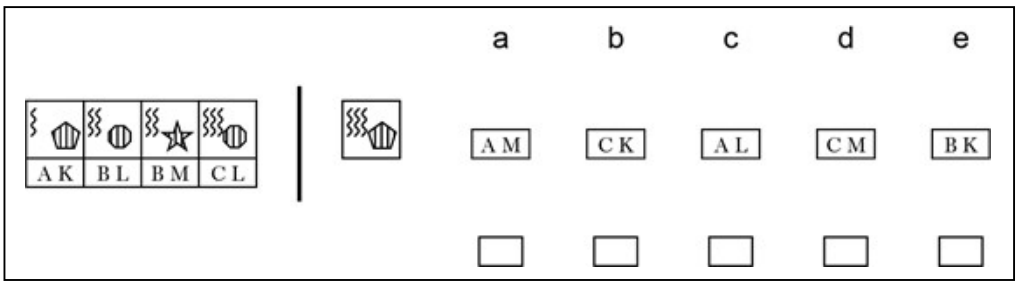
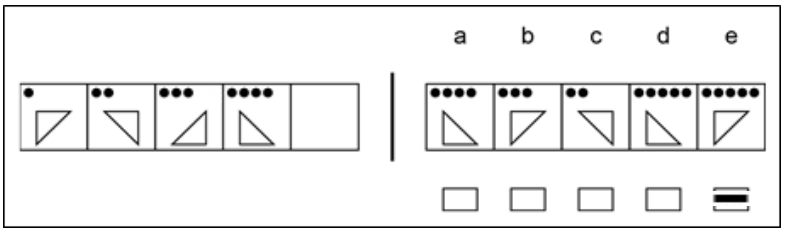


5. MISSING SEQUENCE / TYPE 5

Select the answer that correctly completes the sequence.

Example



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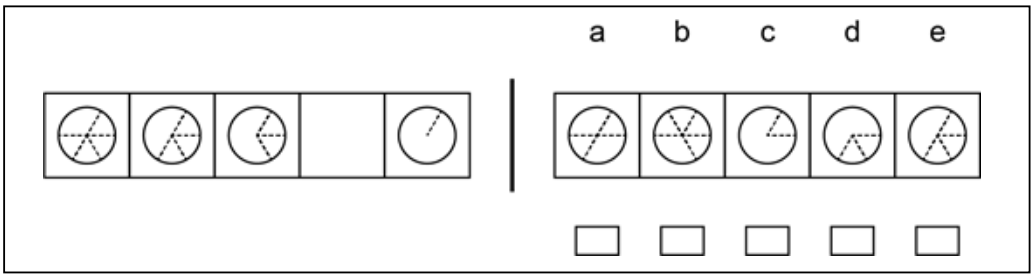

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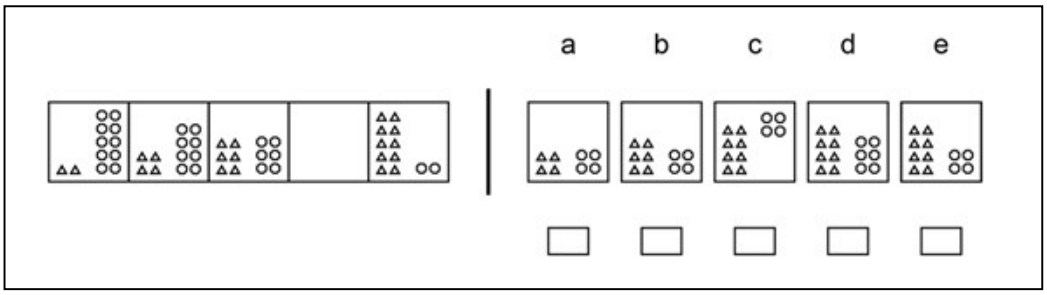
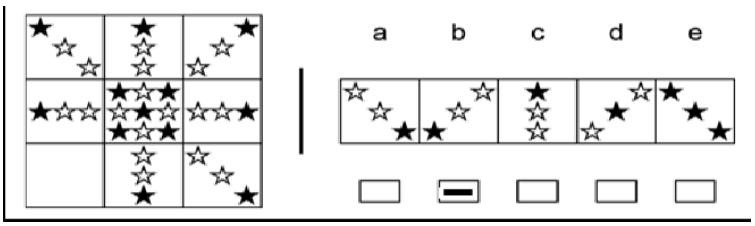


6. MISSING SQUARE / TYPE 6

Select the answer that correctly completes the grid.

Example



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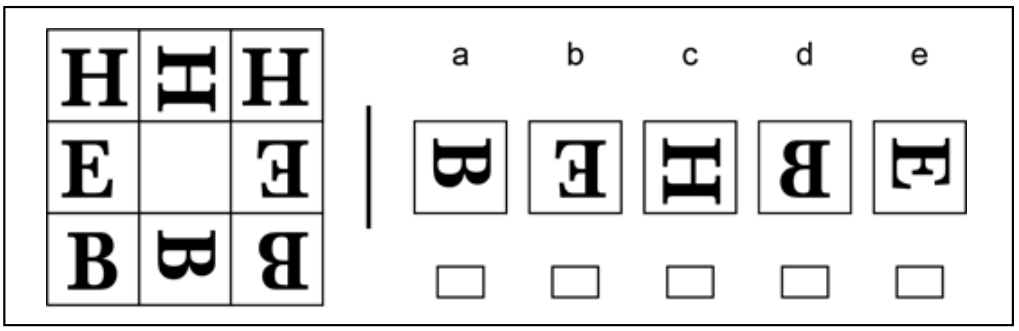

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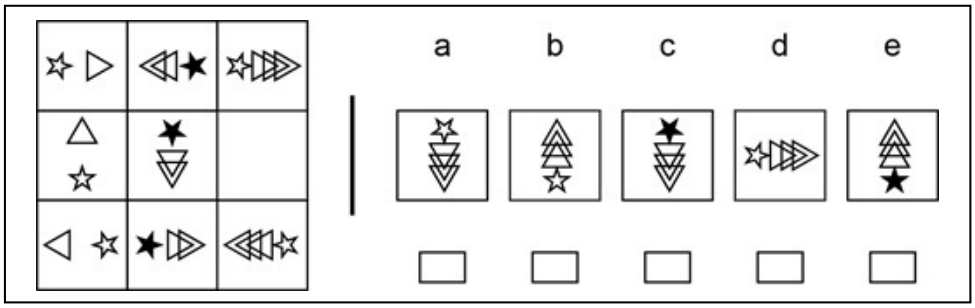
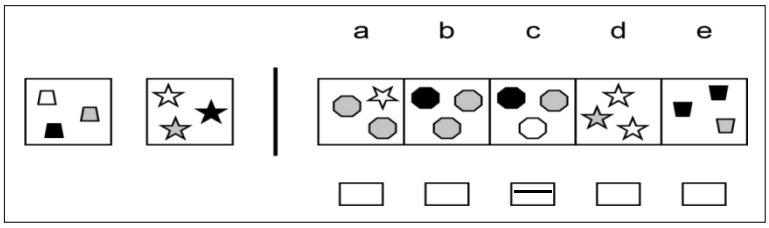


7. MOST LIKE / TYPE 7

Select the answer that is most similar to the two given shapes.

Example



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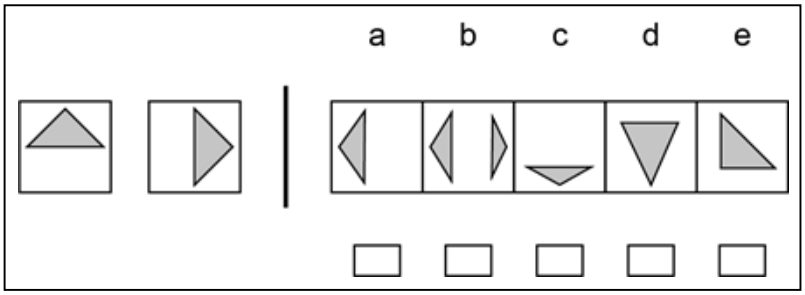

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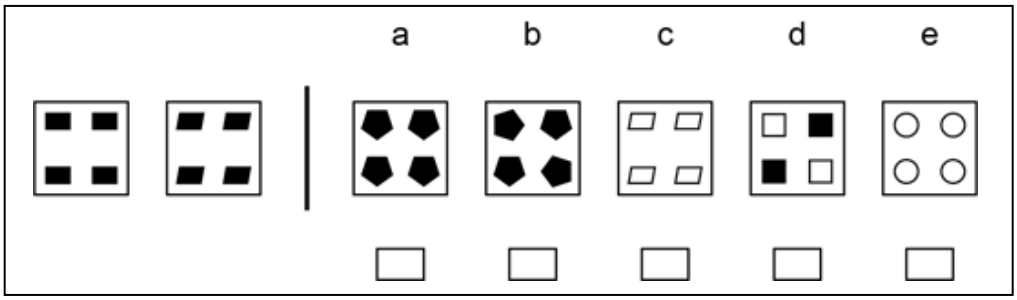

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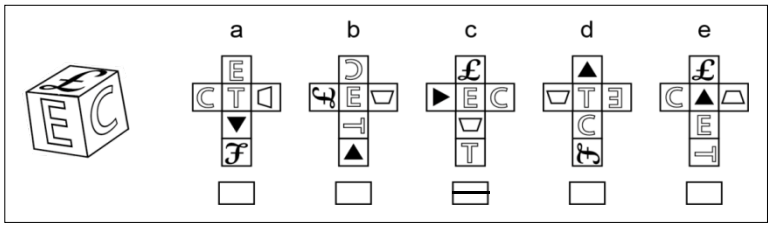
8. NETS / TYPE 8

Select the cube that is made from the given net, or net that makes the given cube.

Example



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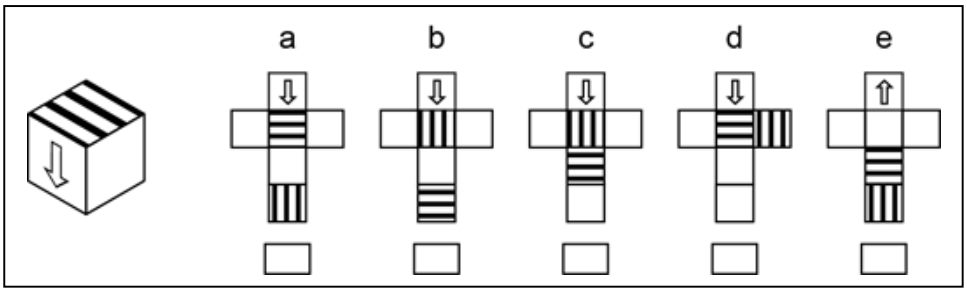
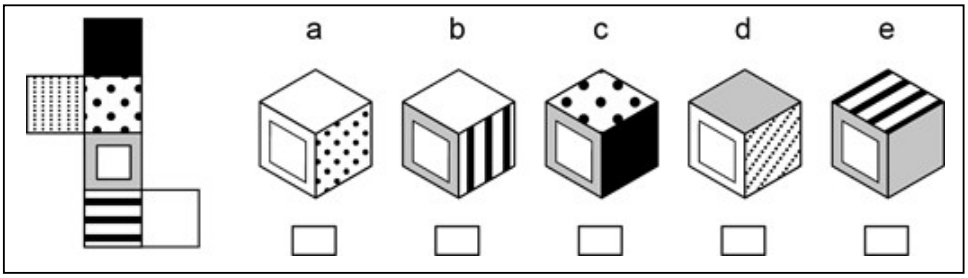


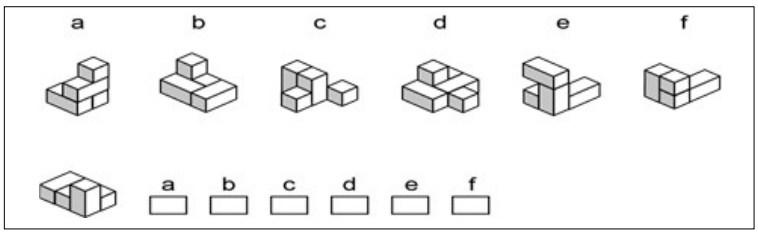
2



9. 3D SHAPES / TYPE 9

You will need to do one of the following:
Select the answer that is a rotation of the given 3D shape; or
Select the building blocks that make the given 3D shape; or
Select the correct 2D view of the given 3D shape.

Example



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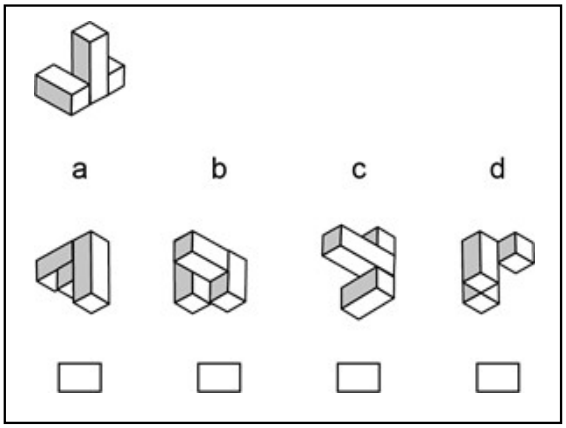
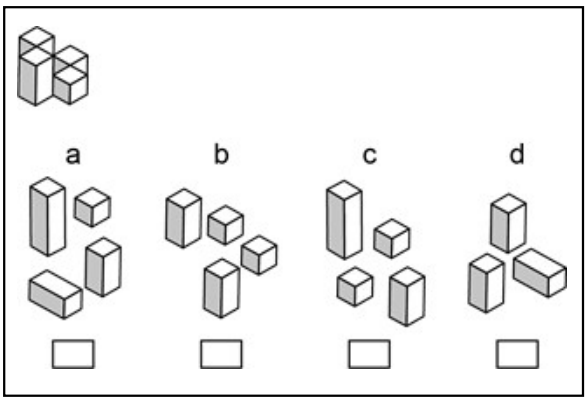


2



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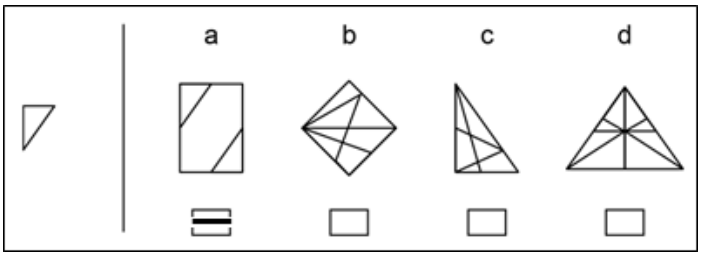
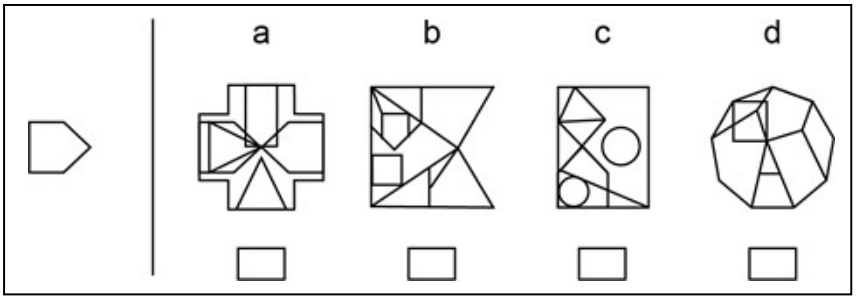
10. HIDDEN SHAPE / TYPE 10

Find the answer option that contains the shape hidden in exactly the same size and orientation.

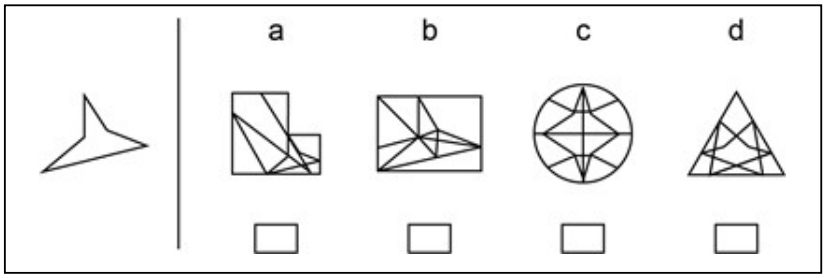
Example



1



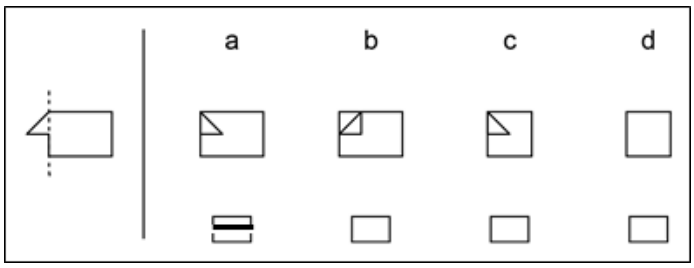
2



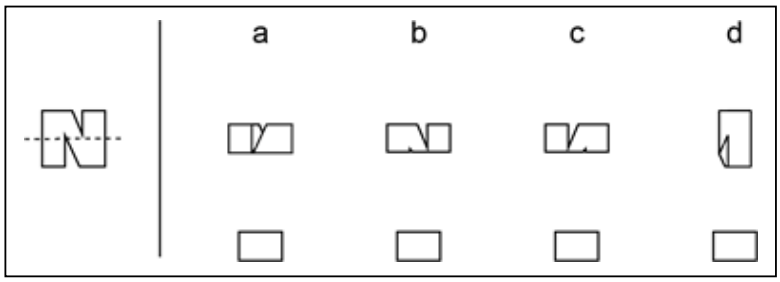
11. FOLD ALONG THE DOTTED LINE / TYPE 11

Select the option that will be formed when the shape is folded along the dotted line.

Example



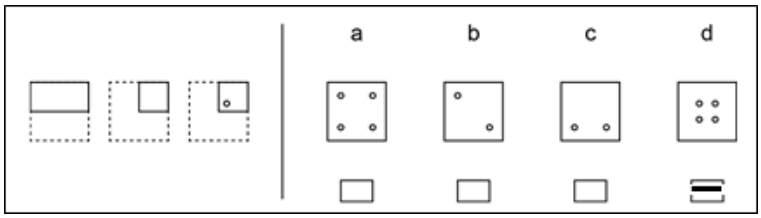
1



12. HOLE PUNCH / TYPE 12

The sequence shows a shape being folded and then hole punched. Select the option that will be created when the hole punched shape is unfolded.

Example



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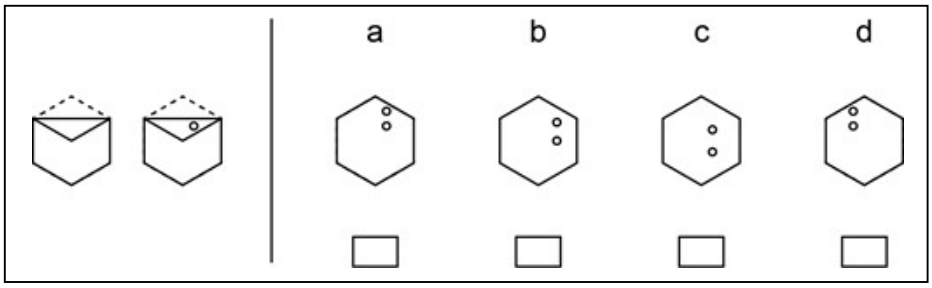
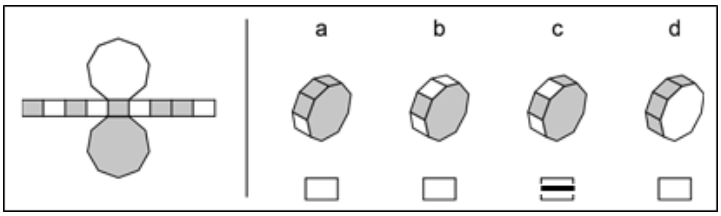


13. SHADED NETS / TYPE 13

Select the 3D shape that is created from the shaded net.

Example



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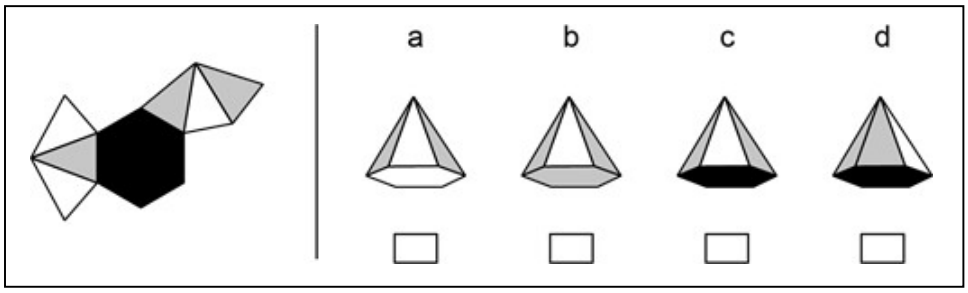


Non-Verbal Reasoning

Like Shapes

1. **E** - (The areas that are white become black and the areas that are black become white.)
2. **D** - (The rectangle is rotated 90 degrees so it is now vertical and not horizontal. One of the letters is rotated 180 degrees and goes at the top of the rectangle while the other stays as it is and goes at the bottom.)

Odd One Out

1. **C** - (All the shapes are centred over the oval except for C)
2. **B** - (In each answer there is an arrow pointing towards a large shape, and on the end of the arrow is a smaller version of the large shape. The odd one out is B as it has a small rhombus pointing towards the octagon and a small octagon pointing towards the rhombus.)

Rotations

1. **C** - (C is a rotation of around 135 degrees clockwise of the original.)
2. **C** - (C is a rotation of around 45 degrees clockwise of the original.)

Code Breaker

1. **D** - (You can work out that V stands for the triangle being black as there are two of them. Therefore, W stands for the triangle having lines through it. All of the top letters are different and the only difference between the three squares is that the triangles are all pointing in different directions. The triangle pointing upwards is therefore O. So, the final answer is OW.)
2. **B** - (You can work out that B stands for there being two wavy lines as there are two of them. Therefore, A stands for there being one wavy line and C stands for there being three wavy lines. In the right hand letters there are two Ls and the similarity between the two is that both of the shapes are octagons. So, K must be the shape being a pentagon. So, the final answer is CK.)

Missing Sequence

1. **C** - (As you move from left to right there is one dashed line taken away from each of the circles. As there is three dashed lines in the third square and one dashed line in the fifth square, there must be two dashed lines in the fourth square. So, the final answer is C.)
2. **E** - (As you move from left to right the number of rows of circles decreases by one while the number of rows of triangles increases by one. So, the final answer must have two rows of circles on the right and four rows of triangles on the left so must be E.)

Missing Square

1. **E** - (In each row, as you move from left to right the letters are rotated 90 degrees clockwise. So the missing square should be a rotation of 90 degrees clockwise of the left hand square in the middle row. The answer is therefore E.)
2. **B** - (As you move across the two completed rows from left to right, a number of things happen: the small star alternates between white and black, the number of triangles increases by one and the entire square is rotated 180 degrees. So the missing square should have a white star, have three triangles and should be pointing upwards. The answer is therefore B.)

Most Like

1. **E** - (In each of the squares on the left there is one large, lightly shaded triangle is the size of about half of the square. The triangles in A and C are both too small, the triangle in D is too big and there are two triangles in B, so the answer has to be E as it is the only lightly shaded triangle that is the size of about half the square.)
2. **A** - (All the shapes are black and all are copies of each other, with no rotation, just in different corners. The only shape that is like this is A as B has some of the shapes rotated.)

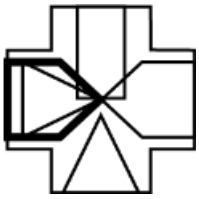
Nets

1. **B** - (You can see from the cube that the base of the arrow has a square with horizontal lines above it. There is also a white square to the right hand side of the arrow. From this information, you can see that the only possible answer is B.)
2. **B** - (From the net you can see that the lightly shaded square with a white square in has a square with vertical lines in to its right hand side. Both of these squares will also have a white square above it when it is folded. From this information, you can see that the answer is B.)

3D Shapes

1. **C** - (The shape is rotated 90 degrees anti-clockwise in the z-axis.)
2. **B** - (The shape must be made from two cuboids of equal height and two cubes of equal height.)

Hidden Shape

1. **A** - 

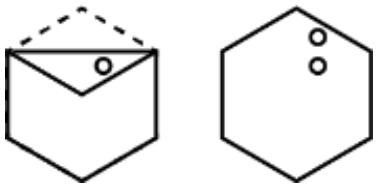
2. **B** -

Fold Along the Dotted Line

1. **A** - (The bottom half of the shape folds up over the top half so that a small part of the top half of the shape is still visible in the background.)

Hole Punch

1. **A** -



Shaded Nets

1. **C** - (You can see from the net that the shape must have a solid black base. You can also see that the colour of the triangular faces must alternate between white and shaded.)